

Autodesk®

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Scaleform

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Autodesk® Scaleform® 4.4

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서문

Autodesk® Scaleform® SDK
PC . Scaleform

XML AS2 XML API . XML XMLNode ,
DOM . XML Scaleform 4.4 .
Scaleform DOM SAX2
XML . Scaleform Player
Expat . XML Windows®, Linux®, MacOS®, Xbox 360®,
PSP® (PlayStation®Portable), PlayStation®2 (PS2), PlayStation®3 (PS3™), Nintendo® Wii™
Scaleform .

Scaleform XML .


```

Ptr<XML::ObjectManager> pObjMgr = *new XML::ObjectManager();
Ptr<XML::Document> pdoc2 = domBuilder.ParseFile("inputfile.xml",
                                                mLoader.GetFileOpener(), pObjMgr);
...

```

ActionScript XML

```

...
#include "GFx/XML/XML_DOM.h"
#include "GFx/XML/XML_Expatriate.h"
using namespace Scaleform;
...
Ptr<XML::Parser> pexpatXmlParser = *new XML::ParserExpatriate;
Ptr<XML::Support> pxmlSupport = *new XML::Support(pexpatXmlParser);
...
//          DOM
XML::DOMBuilder domBuilder(pxmlSupport);
//          DOM
XML::DOMBuilder domBuilder2(pxmlSupport, true);
...
//
...

```

1.3 C++ DOM

```

        DOM (          Scaleform
    ).

```

```

...
using namespace Scaleform;
...
void PrintDOMTree(XML::Node* proot)
{
    switch (proot->Type)
    {
        case XML::ElementNodeType:
        {
            XML::ElementNode* pnode =
                static_cast< XML::ElementNode*>(proot);
            if (pnode->Prefix.GetSize() > 0)
            {
                printf("ELEM - '%s:%s' ns:'%s' prefix:'%s'"
                    " localname: '%s'",
                    pnode->Prefix.ToCStr(),
                    pnode->Value.ToCStr(),

```

```

        pnode->Namespace.ToCStr(),
        pnode->Prefix.ToCStr(),
        pnode->Value.ToCStr());
    }
    else
    {
        printf("ELEM - '%s' ns: '%s' prefix: '"
            " localname: '%s'",
            pnode->Value.ToCStr(),
            pnode->Namespace.ToCStr(),
            pnode->Value.ToCStr());
    }
    for (XML::Attribute* attr = pnode->FirstAttribute;
        attr != NULL; attr = attr->Next)
    {
        printf(" {%s,%s}", attr->Name.ToCStr(),
            attr->Value.ToCStr());
    }
    printf("\n");
    for (XML::Node* child = pnode->FirstChild; child != NULL;
        child = child->NextSibling)
        PrintDOMTree(child);
    break;
}
case TextNodeType:
{
    printf("TEXT - '%s'\n", pnode->Value.ToCStr());
    break;
}
default:
{
    printf("UNKN\n");
}
}
}
...
Ptr<XML::Document> pdoc = domBuilder.ParseFile("inputfile.xml",
                                                Loader.GetFileOpener());
PrintDOMTree(root);

```


1.4

Scaleform

XML

DOM

XML_DOM.cpp

/

```
//
```

```
#ifdef SF_BUILD_DEBUG
```

```
//
```

```
//                                     DOM
```

```
//#define SF_XML_DOCBUILDER_TRACE
```

```
//
```

```
//          DOM
```

```
/(      :                               )
```

```
//#define SF_XML_DOCBUILDER_DOM_TREE_DUMP
```

```
//
```

```
#endif // #ifdef SF_BUILD_DEBUG
```

2. 커스텀 파서의 구현

Scaleform expat .
 XML Gfx::XML::Parser
 Scaleform XML XML_Support.h . XML_Parser.h
 Gfx::XML::ParserHandler SAX2 .

```
namespace Scaleform { namespace Gfx { namespace XML {
//
//      XML
//
//      /
//
//      .
//
// XML
//
class Parser : public RefCountBaseNTS<Parser, Stat_Default_Mem>
{
public:
    virtual ~Parser() {}

    // Parse methods

    virtual bool    ParseFile(const char* pfilename, FileOpenerBase* pfo,
                             ParserHandler* pphandler) = 0;
    virtual bool    ParseString(const char* pdata, UPInt len,
                               ParserHandler* pphandler) = 0;
};
}}} //SF::Gfx::XML
```

XML ParseFile ParseString DOM
 Gfx::XML::ParserLocator .
 Gfx::XML::ParserHandler .

```
namespace Scaleform { namespace Gfx { namespace XML {
//
// SAX2
//
// DOM      SAX2
//
//
//
```

```

//
class ParserHandler : public RefCountBase<ParserHandler, StatMV_XML_Mem>
{
public:
    ParserHandler() { }
    virtual ~ParserHandler() {}

    // 문서의 시작과 끝
    virtual void      StartDocument() = 0;
    virtual void      EndDocument() = 0;

    //
    virtual void      StartElement(const StringRef& prefix,
                                   const StringRef& localname,
                                   const ParserAttributes& atts) = 0;
    virtual void      EndElement(const StringRef& prefix,
                                 const StringRef& localname) = 0;

    //
    virtual void      PrefixMapping(const StringRef& prefix,
                                    const StringRef& uri) = 0;

    //
    virtual void      Characters(const StringRef& text) = 0;

    //
    virtual void      IgnorableWhitespace(const StringRef& ws) = 0;

    //
    virtual void      SkippedEntity(const StringRef& name) = 0;

    // GfX::XML::Parser
    //           . GfX::XML::ParserHandler
    //
    // xmlversion

    virtual void      SetDocumentLocator(const ParserLocator* plocator) = 0;

    //
    virtual void      Comment(const StringRef& text) = 0;

    // ErrorHandler Callbacks
    virtual void      Error(const ParserException& exception) = 0;
    virtual void      FatalError(const ParserException& exception) = 0;
    virtual void      Warning(const ParserException& exception) = 0;
};
}}} //SF::GfX::XML

```

3. 알려진 제한사항

3.1 *ActionScript* XML

ActionScript XMLNode (XML)



```
// DOM 노드에서 참조 획득
XMLNode temp = root.firstChild;
// 노드에 커스텀 속성 설정
temp.someProperty = someValue;
// 참조 드롭 (모든 참조)
XMLNode temp = temp.nextSibling;
// 참조 가져오기
temp = temp.prevSibling;
// 커스텀 속성 찾기
trace(temp.someProperty)
```

someValue Scaleform XMLNode

: XMLNode

ActionScript XMLNode

```
// DOM 노드에서 참조 획득
XMLNode temp = root.firstChild;
// 커스텀 속성을 노드에 설정
temp.attributes.someProperty = someValue;
// 참조 드롭 (모든 참조)
```

```
XMLNode temp = temp.nextSibling;
// 참조 가져오기
temp = temp.prevSibling;
// 커스텀 속성 찾기
trace(temp.attributes.someProperty)
```

출력값은 Scaleform 내에서 someValue 가 된다.

3.2 *ActionScript XML.status*

ActionScript 2.0 XML

ActionScript

- 0 ,
- -2 CDATA
- -3 XML
- -4 DOCTYPE
- -5
- -6 XML
- -7
- -8
- -9
- -10

ActionScript

Expat

```
//
// XML_ERROR_NONE = 0
// XML_ERROR_INVALID_TOKEN = 0 (ie: XML.parse("http://www.google.com"))
// XML_ERROR_UNCLOSED_CDATA_SECTION = -2
// XML_ERROR_UNCLOSED_TOKEN = -3
// XML_ERROR_INVALID_TOKEN = -4
// XML_ERROR_INVALID_TOKEN = -5
// XML_ERROR_UNCLOSED_TOKEN = -8
// XML_ERROR_NO_ELEMENTS = -9
// XML_ERROR_TAG_MISMATCH = -10
//
```

		1	1	.
GfX::XML::DOMBuilder	XML			GfX::XML::ParserLocator
				.

3.3 ActionScript XML.doc TypeDecl

	"ActionScript XML	.	DOCTYPE
XML.docTypeDecl	DTD	.	"
Scaleform		.	